No.	Status	Date
HIS-2011-0207-DS-6-SG-101	Issued	2011-02-18

UT Thickness Data Sheet - 6-SG-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 6-SG-101, located within the ATU-100. The scope of work involves the recording of wall thickness to monitor the integrity of the sour gas inlet piping.

Kevin Doucette, acting as the lead UT technician and SNT-TC-1A UT Level II for Halward Inspection Services Ltd., is responsible for the execution of the measurements and the recording of the data.

Technical Specifications

The subject line, 6-SG-101, is constructed from A106 Gr.B with an outer diameter of 168.3 mm and a nominal thickness of 7.11 mm. The design limits for this line are 1965 kPa(g) and 66 °C.

The minimum allowable thickness (t-min) is established at 3.56 mm. Any reading recorded below this threshold is subject to an NCR.

Procedural Compliance

Thickness measurements were performed in accordance with HIS-UT-002 Rev 1 and the site-specific requirements of QSCP-NDT-001 Rev 2. These procedures ensure compliance with the CUI program, including requirements for insulation removal and inspection ports.

The ultrasonic testing was conducted following the guidelines set forth in ASTM E797.

Inspection Results and Analysis

The data collected for 6-SG-101 is integrated into the findings of HIS-2011-0207. This 5-year Class 1 survey identifies the current state of the sour gas and rich amine circuits.

Based on the analysis in HIS-2011-0207, the rich amine circuit is currently at half remaining life.

Thickness readings, measured 2011-02-18

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2006-02-24	6.59	6.39	0.049	0.040	>50	2016-02-18	
6-SG-101-02	7.11	3.56	2006-02-24	6.51	6.14	0.061	0.074	34.7	2016-02-18	
6-SG-101-03	7.11	3.56	2006-02-24	6.42	6.17	0.061	0.050	42.9	2016-02-18	
6-SG-101-04	7.11	3.56	2006-02-24	6.28	6.05	0.070	0.046	35.8	2016-02-18	
6-SG-101-05	7.11	3.56	2006-02-24	6.22	5.87	0.085	0.070	27.2	2016-02-18	
6-SG-101-06	7.11	3.56	2006-02-24	6.49	6.27	0.054	0.044	>50	2016-02-18	
6-SG-101-08	7.11	3.56	—	—	6.33	—	—	—	2016-02-18	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2006-02-24	6.26	5.75	0.093	0.102	21.4	2016-02-18	
6-SG-102-02	7.11	3.56	2006-02-24	6.25	5.80	0.088	0.090	24.8	2016-02-18	
6-SG-102-03	7.11	3.56	2006-02-24	6.26	5.80	0.083	0.092	24.3	2016-02-18	
6-SG-102-04	7.11	3.56	2006-02-24	6.22	5.73	0.084	0.098	22.1	2016-02-18	
6-SG-102-05	7.11	3.56	2006-02-24	6.72	6.46	0.045	0.052	>50	2016-02-18	
6-SG-102-06	7.11	3.56	—	—	6.41	—	—	—	2016-02-18	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2006-02-24	8.27	8.32	0.013	-0.010	>50	2021-02-18	
4-RA-101-02	8.56	3.81	2006-02-24	8.27	8.20	0.023	0.014	>50	2021-02-18	
4-RA-101-03	8.56	3.81	2006-02-24	8.10	7.98	0.042	0.024	>50	2021-02-18	
4-RA-101-04	8.56	3.81	2006-02-24	8.34	8.21	0.028	0.026	>50	2021-02-18	
4-RA-101-05	8.56	3.81	2006-02-24	8.29	8.16	0.029	0.026	>50	2021-02-18	
4-RA-101-07	8.56	3.81	2006-02-24	8.30	8.30	0.020	0.000	>50	2021-02-18	
4-RA-101-08	8.56	3.81	2006-02-24	8.07	8.02	0.044	0.010	>50	2021-02-18	
4-RA-101-09	8.56	3.81	2006-02-24	8.43	8.32	0.017	0.022	>50	2021-02-18	
4-RA-101-10	8.56	3.81	2006-02-24	8.27	8.28	0.026	-0.002	>50	2021-02-18	
4-RA-101-11	8.56	3.81	2006-02-24	8.14	8.04	0.043	0.020	>50	2021-02-18	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2006-02-24	8.41	8.33	0.014	0.016	>50	2021-02-18	
4-RA-102-02	8.56	3.81	2006-02-24	8.15	7.99	0.039	0.032	>50	2021-02-18	
4-RA-102-03	8.56	3.81	2006-02-24	8.27	8.04	0.035	0.046	>50	2021-02-18	
4-RA-102-04	8.56	3.81	2006-02-24	8.24	7.96	0.039	0.056	>50	2021-02-18	
4-RA-102-05	8.56	3.81	2006-02-24	8.19	8.07	0.032	0.024	>50	2021-02-18	
4-RA-102-06	8.56	3.81	2006-02-24	8.15	8.01	0.033	0.028	>50	2021-02-18	
4-RA-102-07	8.56	3.81	2006-02-24	8.29	8.10	0.032	0.038	>50	2021-02-18	
4-RA-102-08	8.56	3.81	2006-02-24	8.05	7.88	0.041	0.034	>50	2021-02-18	
4-RA-102-09	8.56	3.81	2006-02-24	8.18	7.98	0.038	0.040	>50	2021-02-18	
4-RA-102-11	8.56	3.81	—	—	8.19	—	—	—	2021-02-18	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2006-02-24	10.76	10.47	0.035	0.058	>50	2021-02-18	
6-RA-103-02	10.97	4.57	2006-02-24	10.50	10.23	0.058	0.054	>50	2021-02-18	
6-RA-103-03	10.97	4.57	2006-02-24	10.67	10.42	0.038	0.050	>50	2021-02-18	
6-RA-103-04	10.97	4.57	2006-02-24	10.40	9.97	0.068	0.086	>50	2021-02-18	
6-RA-103-05	10.97	4.57	2006-02-24	10.34	10.10	0.057	0.048	>50	2021-02-18	
6-RA-103-06	10.97	4.57	2006-02-24	10.74	10.56	0.030	0.036	>50	2021-02-18	
6-RA-103-07	10.97	4.57	2006-02-24	10.66	10.55	0.029	0.022	>50	2021-02-18	
6-RA-103-08	10.97	4.57	2006-02-24	10.58	10.44	0.035	0.028	>50	2021-02-18	
6-RA-103-09	10.97	4.57	2006-02-24	10.67	10.56	0.030	0.022	>50	2021-02-18	
6-RA-103-10	10.97	4.57	2006-02-24	10.27	10.05	0.065	0.044	>50	2021-02-18	
6-RA-103-13	10.97	4.57	—	—	10.45	—	—	—	2021-02-18	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2006-02-24	8.16	7.95	0.037	0.042	>50	2021-02-18	
4-RA-104-02	8.56	3.81	2006-02-24	8.23	8.14	0.027	0.018	>50	2021-02-18	
4-RA-104-03	8.56	3.81	2010-09-13	8.59	8.56	0.069	0.069	>50	2021-02-18	
4-RA-104-04	8.56	3.81	2010-09-13	8.52	8.60	-0.185	-0.185	>50	2021-02-18	
4-RA-104-05	8.56	3.81	2010-09-13	8.43	8.57	-0.324	-0.324	>50	2021-02-18	
4-RA-104-06	8.56	3.81	2006-02-24	8.09	7.98	0.041	0.022	>50	2021-02-18	
4-RA-104-07	8.56	3.81	2006-02-24	7.97	7.78	0.054	0.038	>50	2021-02-18	
4-RA-104-08	8.56	3.81	—	—	8.18	—	—	—	2021-02-18	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2011-02-18	_____